

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

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COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I Narra Sai Mounika Reg no: 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Sai Mounika
Signature of the student:

Narra Sai Mounika
Name of the student:

QUESTIONS:05

Total Marks:25

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Answer:

- Gradient Descent updates model parameters to minimize the loss function.
- A high initial learning rate allows the model to move quickly toward the optimum region.
- Large updates help escape shallow local minima and flat regions.
- Gradually decreasing the learning rate enables smaller and more precise updates near the optimum.
- Learning rate decay often leads to faster convergence than a fixed learning rate.
- This strategy performs better when the dataset is large and the loss function is smooth.
- It combines rapid initial learning with accurate final optimization.
- However, if the initial learning rate is extremely high, the algorithm may overshoot the minimum.
- Excessive overshooting can cause oscillations or divergence.
- Therefore, a high learning rate followed by decay provides a good balance between training speed and convergence accuracy.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

Answer:

For a Gaussian distribution with known variance:

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

- The MLE of the mean is the sample average.
- It is obtained by maximizing the likelihood of observing the given data.
- When outliers are present, the sample mean shifts toward extreme values.
- Even a few outliers can significantly affect the estimate.
- Therefore, MLE is sensitive to outliers and is not robust.
- As sample size increases, the estimate becomes more stable.
- The variance of the estimator decreases with more observations.
- The estimated mean approaches the true population mean.
- This property is known as consistency.
- Hence, MLE performs very well on clean data but poorly when many outliers are present.

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

Answer:

- The dataset contains 10,000 features but only 500 samples.
- This creates a high-dimensional learning problem.
- The model may memorize training data and overfit.
- Training becomes computationally expensive.
- First, standardize all features to a common scale.
- Next, perform feature selection using LASSO or Mutual Information.
- Apply PCA to reduce dimensionality while preserving important information.
- Use regularized models such as Logistic Regression with L1/L2 regularization.
- Validate performance using cross-validation.
- Feature selection, PCA, and regularization reduce model complexity and improve generalization, thereby preventing overfitting.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

Answer:

- Hidden neurons learn intermediate representations of the input data.
- Increasing the number of hidden neurons increases model capacity.

- More neurons allow the network to capture complex nonlinear relationships.
- The network can represent more complicated decision boundaries.
- According to the Universal Approximation Theorem, larger networks can approximate complex functions.
- However, more neurons also increase the number of trainable parameters.
- Excessive parameters may cause the network to memorize training data.
- This leads to overfitting and poor test performance.
- Generalization decreases when the model learns noise instead of useful patterns.
- Therefore, the number of neurons should be chosen carefully to balance learning capacity and generalization.

5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

Answer:

- High-dimensional datasets contain a very large number of features.
- During backpropagation, gradients must be computed for many parameters.
- Some gradients become noisy or insignificant.
- SGD requires more iterations to find optimal parameters.
- Convergence becomes slower due to the large search space.
- High-dimensional data is often sparse, making learning difficult.
- The model may fit noise and overfit the training data.
- Generalization performance on unseen data decreases.
- PCA or Autoencoders can reduce dimensionality and remove redundant features.
- Regularization methods such as L1, L2, Dropout, and Early Stopping reduce overfitting and improve model performance.